

# Project Report

---

## Task 8: Web Scraper (CLI)

Developer: Harsh Bhanarkar

### 1. Objective

To develop a command-line application that retrieves and analyzes publicly accessible web pages. The application extracts useful information such as the page title, meta description, headings, links, and images, and stores the extracted data in a text file.

### 2. Project Overview

The Web Scraper is a modular Python CLI application that accepts a website URL, downloads the HTML content using HTTP requests, parses the document with BeautifulSoup, displays the extracted information, and saves the results for future reference.

### 3. Technologies Used

- Python 3
- requests
- beautifulsoup4 (BeautifulSoup)
- pathlib

### 4. Functional Modules

- main.py – Controls application flow.
- scraper.py – Fetches website HTML.
- parser.py – Extracts page information and saves output.
- menu.py – Displays menus.
- utils.py – Validates user input.

### 5. Workflow

1. Launch the application.
2. Enter a valid website URL.
3. Download HTML content.
4. Parse the document.
5. Extract title, description, headings, links and images.
6. Display the results.
7. Save the extracted data to output/scraped\_data.txt.

## 6. Key Features

- Extract page title
- Extract meta description
- Count links and images
- Display H1/H2 headings
- Save extracted information
- Handle invalid URLs and connection errors

## 7. Error Handling

- URL validation
- Connection error handling
- Timeout handling
- HTTP error handling

## 8. Project Structure

```
assets/  
docs/  
modules/  
output/  
main.py  
requirements.txt  
README.md
```

## 9. Testing

The application was tested using websites such as <https://example.com>, <https://www.python.org>, and <https://github.com>. Invalid URLs and network failures were also tested.

## 10. Conclusion

The project demonstrates practical implementation of web scraping, HTML parsing, file handling, and modular Python application development.

## 11. Future Enhancements

- Export to CSV/JSON
- Crawl multiple pages
- Extract tables
- Keyword search
- Generate PDF reports